

Lonely Chess

A chess-based esoteric language

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Purpose

A chess-based esoteric language in which the source code is a standard Portable Game Notation text format.

Target Domain

Recreational coding and exploring the possibility of using the paradigm of chess for rudimentary programming.

Design

Syntax & Semantics

The language operates via **State-Based Instantiation**. The board state acts as the register, and specific pieces of coordinates trigger specific logic gates

- Example:
 - Using White's Queen's side Knight to instantiate
 - Nb1 to Na3: **Integer** Mode
 - Nb1 to Nc3: **String** Mode
 - Variable Naming
 - Using the **Address** of the White's pawn as the name of a variable
 - The Rook Register:
 - The Black Rook on a8 acts as the **Write Head**. Moving the Rook across the 6th rank (b6 to h6) flips bits to create binary values, according to the ASCII table.
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Control Structures

Following along the **State-Based** structure, by moving a certain piece, it will do accordingly to what we wanted to do. All the control functions will be based on the white pieces.

- If Statements
 - Using White's King's side Knight to activate the if statement and moving the black pawn to access which variable to compare and set its conditions accordingly.
- Loops
 - By making space for White's King, we can move the King from e1 to e2 to start a for loop
- Print Function
 - Using White's Queen to move to a certain tile, move the variable pawn that we wanted, and then move the Queen back to solidify the print statement.

Ecosystem

An interpreter that reads a sequence of chess moves and executes them as a program (e.g., producing outputs like FizzBuzz or Hello World)

A visual debugger that replays the game move-by-move to show how variables and values change on the board

A translator that converts chess notation into a readable pseudo-code format to help users understand program logic

A standard library of predefined move sequences for common operations like variable initialization, loops, conditionals, and printing

An interactive sandbox or IDE where users can write, test, and debug programs directly on a virtual chessboard

Support for importing and exporting PGN files, so real or generated games can be used as programs

Challenges

Simple operations require many moves, making programs long and inefficient

The turn-based nature forces alternating moves, requiring extra “waiting moves” that do not contribute to computation

The meaning of moves can depend on context, making interpretation and parsing more complex

The chessboard has limited space, restricting how much data and how many variables can be represented at once

Debugging is difficult because logic is spread across many moves rather than clearly structured code

Execution is slow compared to traditional programming due to the large number of required moves